

## Quiz 1 Study Guide

*The quiz will be in-class on Friday, March 13. It will be 15 minutes long. Some combination of the following problems will be on the quiz. You do not need to turn this study guide in; it is purely for your benefit. On the exam, please aim to answer the questions as precisely as you can (when grading your answers, I will be looking for mathematical definitions, not examples).*

### 1 Questions from before quiz 1 (some of these will be on quiz 2)

**Problem 1.** What does it mean for two events  $A$  and  $B$  to be **mutually exclusive**?

**Problem 2.** In order for a function  $\mathbb{P}$  to be a **probability measure**, it must satisfy three axioms. State all three axioms.

**Problem 3.** Let  $A$  and  $B$  be events, and assume that  $\mathbb{P}[A] > 0$ . What is the **conditional probability of  $B$  given  $A$** ?

**Problem 4.** What is a  **$k$ -tuple**?

**Problem 5.** Describe the difference between a **combination** and a **permutation**.

**Problem 6.** A **set** is an \_\_\_\_\_ collection of elements, all of which are \_\_\_\_\_.

**Problem 7.** What does it mean for two events to be **disjoint**?

**Problem 8.** What is the **sample space** of an experiment? What is an **event**?

**Problem 9.** What does it mean for a sequence of events  $E_1, E_2, E_3, \dots$  to be **pairwise disjoint**?

**Problem 10.** Let  $A$  and  $B$  be events. Define the following terms

- the **complement** of  $A$
- the **union** of  $A$  and  $B$
- the **intersection** of  $A$  and  $B$

**Problem 11.** What does it mean for two events  $A$  and  $B$  to have a **positive relationship**? A **negative relationship**?

### 2 New stuff since quiz 1

**Problem 12.** What does it mean for  $n$  events  $A_1, \dots, A_n$  to be **mutually exhaustive**?

**Problem 13.** What does it mean for two events to be **independent**? What does it mean for them to be **dependent**?

**Problem 14.** What is a **random variable**?

**Problem 15.** What does it mean for  $n$  events  $A_1, \dots, A_n$  to be **mutually independent**?

**Problem 16.** What does it mean for a random variable to be **discrete**?

**Problem 17.** What is the **probability mass function** (pmf) of a discrete random variable?

**Problem 18.** What is the **cumulative distribution function** (cdf) of a random variable  $X$ ?

**Problem 19.** Suppose  $X$  is a discrete random variable. What is the **expected value** of  $X$ ?

**Problem 20.** What is **linearity of expectation**?

**Problem 21.** What is the **variance** of a random variable? The **standard deviation**?

- Problem 22.** What is a *binomial* random variable? Make sure to state the probability mass function.
- Problem 23.** What is a *geometric* random variable? Make sure to state the probability mass function.
- Problem 24.** What is a *Poisson* random variable? Make sure to state the probability mass function.
- Problem 25.** What is a *probability density function*?
- Problem 26.** What does it mean for a random variable to be *continuous*?
- Problem 27.** What is a *uniform* random variable? Make sure to write down the pdf.
- Problem 28.** What is an *exponential* random variable? Make sure to write down the pdf.
- Problem 29.** What is the *expected value* of a continuous random variable?